



FNOReg: Resolution-Robust Medical Image Registration Method Based on Fourier Neural Operator

Nikita A. Drozdov  and Dmitry V. Sorokin  

Laboratory of Mathematical Methods of Image Processing, Faculty of Computational Mathematics and Cybernetics, Lomonosov Moscow State University, Moscow, Russia
drozdovna@my.msu.ru, dsorokin@cs.msu.ru

Abstract. Medical image registration plays a crucial role in diagnosis, treatment planning, and anatomical studies. Classical methods, relying on iterative optimization algorithms, are complex and computationally intensive. Recent advances in deep learning, particularly with convolutional neural networks (CNNs) like VoxelMorph, have shown promise. However, they often yield non-smooth deformation fields and require inference at the same image resolution as the training data. To overcome these challenges, we introduce FNOReg, a novel model based on Fourier Neural Operators (FNOs), which can be trained on reduced-resolution images without quality loss and produces smoother deformation fields. We evaluated FNOReg on 2D and 3D datasets, demonstrating comparable quality to popular models like VoxelMorph, Fourier-Net, and TransMorph when trained at the same resolution. However, these models exhibit significant quality decreases of up to 24.9% for 2D and 24.6% for 3D data, when trained at halved resolutions. In contrast, FNOReg demonstrates only marginal quality decreases of up to 0.8% for 2D and 2.7% for 3D data. This flexibility is essential for efficiently handling large image resolutions, particularly in 3D imaging. Moreover, FNOReg produces smoother deformation fields. The code is available at <https://github.com/anac0der/fnoreg>.

Keywords: Biomedical image registration · Fourier Neural Operator · unsupervised learning

1 Introduction

Image registration is one of the key tasks in the field of image processing and is widely used in the analysis of medical images. Among the many applications of image registration, one can distinguish: (i) Combining information obtained using various imaging devices or protocols to facilitate diagnosis and treatment

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planning. (ii) Studies that examine structural or anatomical changes in the same areas of the body. Registration can be classified based on various factors, including the dimensionality of the data (for example, 2D, 3D), the modality of the image (for example, computer tomography or MRI), objects in the image (for example, brain, lungs, and heart), and the form of transformation (rigid or non-rigid registration). The goal of algorithms for non-rigid registration is to predict a deformation field for the registered (moved) image so that the difference between the fixed and moved images is minimal. Despite the simple idea, the task turns out to be ill-posed and, therefore, difficult in practice. The solution to such a problem may not be unique [19]. In practice, some constraints are imposed on the deformation field, such as smoothness, symmetry, or diffeomorphism. These constraints serve as regularization for an iterative algorithm that solves the problem of minimizing the difference between images.

Before the deep learning era, non-rigid registration was performed using iterative optimization algorithms. Methods based on this approach include Free-Form Deformation [22], LDDMM [5], Demons [23], Elastix [13], ANTs [3], NiftyReg [18], and Flash [25]. These methods are widely used and have a mathematical foundation but also require the selection of parameters for each pair of images and significant computational resources, which limits their use for the registration of large images in real-time. Currently, there is an increase in the number of approaches to the registration of biomedical images based on deep learning. Their advantage is the fast model inference without the need to select parameters for each pair of images. The most effective and popular methods, such as VoxelMorph [4], predict the deformation field using a convolutional auto-encoder (for example, U-Net [21]). Such methods are faster than classical ones since they only require forward propagation of a pair of images through the neural network and can be run on GPU out of the box. After the success of VoxelMorph, other neural network-based registration methods have appeared, in one way or another, improving the model which predicts the deformation field. For example, in [6], the authors replaced the convolutional blocks of the encoder with more complex transformer blocks with an attention mechanism, thus increasing model performance on medical datasets. However, neural network-based models often encounter issues with non-smooth deformation fields and require additional regularization in the loss function. A recent model called Fourier-Net [11] aims to address this problem by predicting a band-limited deformation field in the Fourier domain, followed by zero-padding and inverse Fourier transform. However, this approach involves low-pass filtering of the deformation field spectra with a rectangular step function, which can introduce various artifacts such as ringing and aliasing. Additionally, all current neural network-based approaches to image registration have a major disadvantage: they rely on classical convolutional layers with small kernels for deep learning, resulting in a degradation of registration quality when input data resolution changes.

Some classical approaches to registration are based on solving the Euler-Lagrange equation for the corresponding loss functional [5]. Typically, partial differential equations (PDEs) are solved using numerical techniques, but in recent

years, physics-informed neural operators (PINOs) [15] are widely used for such tasks. One of the PINO architectures is called the Fourier Neural Operator (FNO) [14] and is designed to solve the parametric PDE. FNO has one important property – since all operations are performed on a fixed set of frequencies in the Fourier space, this architecture is capable of modeling non-local convolution kernels. Consequently, FNO is weakly sensitive to the shape of input data and can be used for image registration at different resolutions. Recently, FNO was used for other image analysis problems such as image segmentation [24] and classification [12].

In this paper, we propose FNOReg – a model for image registration based on the original FNO architecture. The main advantage of this approach is its robustness to input data resolution. Our model has two significant improvements over the original FNO: we incorporated feature extractors based on the Spectral Transform layer [7] and used additional residual connections in the Fourier layers. Additionally, we tailored the training pipeline for the image registration problem. We evaluated FNO and FNOReg on the OASIS-1 dataset [16] for 2D and 3D data, and our model achieved similar quality compared to popular models widely used for image registration (VoxelMorph [4], TransMorph [6], FourierNet [11]). However, the performance of our FNOReg models does not significantly degrade when trained and evaluated on data with reduced resolutions, whereas other models exhibit a significant decrease in registration quality under these conditions. This feature of the method is important for reducing memory consumption when working with large data and can be especially useful for 3D data. Moreover, we compared the spectra of deformations obtained from different models and demonstrated that FNO and FNOReg provide smoother deformation fields compared to other models.

2 Methodology

2.1 From Image Registration to a System of PDEs

Let us define a spatial domain $\Omega \subset \mathbb{R}^2$ (we consider the two-dimensional case for simplicity, but all calculations can be extended to the case of an arbitrary dimension). We define $\mathbf{x} = (x_1, x_2) \in \Omega$ to be an arbitrary point in domain Ω and $\boldsymbol{\phi}(\mathbf{x}) = (\phi_1(\mathbf{x}), \phi_2(\mathbf{x}))$ to be a displacement field representing the image deformation. Let $f(\mathbf{x})$ be a continuous function in Ω with continuous first order partial derivatives on Ω . This function corresponds to the fixed image while similarly introduced functions $m(\mathbf{x})$ and $m_\phi(\mathbf{x})$ corresponds to the moving and moved images accordingly. Next, we demonstrate that the image registration problem can be reduced to a parametric system of partial differential equations that can be subsequently solved using neural operators.

Let $\Omega_p \subset \Omega$ be a finite set of points from Ω which is a pixel grid for discrete versions of fixed, moving and moved images. Then, the common loss functional for image registration task in discrete case can be written as:

$$L_{f,m}(\boldsymbol{\phi}) = \frac{1}{|\Omega_p|} \sum_{\mathbf{p} \in \Omega_p} D_{f,m_\phi}(\mathbf{p}) + \frac{\lambda}{|\Omega_p|} \sum_{\mathbf{p} \in \Omega_p} \|\nabla \boldsymbol{\phi}(\mathbf{p})\|^2,$$

where $D_{f,m_\phi}(\mathbf{p})$ is a pixel-wise difference between two images (for example, $D_{f,m}(\mathbf{p}) = (f(\mathbf{p}) - m(\mathbf{p}))^2$), the second term is the regularization term, and λ is the regularization parameter.

In continuous case, we can rewrite the loss functional in an integral form as

$$\hat{L}_{f,m}(\phi) = \int_{\Omega} \left[D_{f,m_\phi} + \lambda \left(\left[\frac{\partial \phi_1}{\partial x_1} \right]^2 + \left[\frac{\partial \phi_1}{\partial x_2} \right]^2 + \left[\frac{\partial \phi_2}{\partial x_1} \right]^2 + \left[\frac{\partial \phi_2}{\partial x_2} \right]^2 \right) \right] dx_1 dx_2.$$

Now, if we denote the expression under integral as $F_{f,m}(\phi)$, and the partial derivatives $\partial \phi_i / \partial x_j$ as $\phi_{i,j}$, we can write the Euler-Lagrange system [8] for $\hat{L}_{f,m}$:

$$\begin{aligned} \frac{\partial F_{f,m}}{\partial \phi_1} - \frac{\partial}{\partial x_1} \left(\frac{\partial F_{f,m}}{\partial \phi_{1,1}} \right) - \frac{\partial}{\partial x_2} \left(\frac{\partial F_{f,m}}{\partial \phi_{1,2}} \right) &= 0; \\ \frac{\partial F_{f,m}}{\partial \phi_2} - \frac{\partial}{\partial x_1} \left(\frac{\partial F_{f,m}}{\partial \phi_{2,1}} \right) - \frac{\partial}{\partial x_2} \left(\frac{\partial F_{f,m}}{\partial \phi_{2,2}} \right) &= 0. \end{aligned} \quad (1)$$

It can be shown that after transformations the system (1) can be expressed as:

$$\frac{\partial D_{f,m_\phi}}{\partial \phi_1} - 2\lambda \cdot \Delta \phi_1 = 0; \quad \frac{\partial D_{f,m_\phi}}{\partial \phi_2} - 2\lambda \cdot \Delta \phi_2 = 0. \quad (2)$$

where $\Delta = \frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2}$ is the Laplace operator. Thus, if the displacement field ϕ_{opt} is the point minimum for the $\hat{L}_{f,m}(\phi)$, then it is the solution of system (2). This system is parameterized by the fixed image f and moving image m and we can apply neural operators to approximate the solution operator of this equation.

2.2 Fourier Neural Operator

Fourier Neural Operator (FNO) is a deep learning model for solving parametric partial differential equations [14]. FNO does not require knowledge of the equation itself and based only on the provided data, which leads to the classical supervised or unsupervised learning task.

The architecture of FNO is inspired by the mathematical principles that lies behind the solution of some types of operator equations. More specifically, if we have an equation $Lu(\mathbf{x}) = f(\mathbf{x})$ where L is a linear differential operator and $\mathbf{x} \in \Omega \subset \mathbb{R}^n$, then the solution $u(\mathbf{x})$ of this equation is given by formula

$$u(\mathbf{x}) = \int_{\Omega} G(\mathbf{x}, \mathbf{s}) f(\mathbf{s}) d\mathbf{s},$$

where $G(\mathbf{x}, \mathbf{s})$ is a Green function of operator L . As proposed in [1, 14], $u(\mathbf{x})$ can be approximated using the following iterative process:

$$u_{t+1}(\mathbf{x}) := \sigma(W_t u_t(\mathbf{x}) + (K u_t)(\mathbf{x})), \quad t = 0, \dots, N-1, \quad (3)$$

where

$$(Ku_t)(\mathbf{x}) := \int_{\Omega} \kappa_t(\mathbf{x} - \mathbf{y}) u_t(\mathbf{y}) d\mathbf{y}. \quad (4)$$

Here $u_t(\mathbf{x}) \in \mathbb{R}^{d_{u_t}}$ is the function of $\mathbf{x}$, $W_t \in \mathbb{R}^{d_{u_t+1} \times d_{u_t}}$ is a learnable linear transformation, $\sigma(\cdot)$ is a nonlinear activation function and $\kappa_t(\mathbf{x} - \mathbf{y})$ is a learnable kernel function of convolutional operator K . We can think about $u_t(\mathbf{x})$ as the output of network hidden layer with d_{u_t} data channels.

We can apply the convolution theorem to the operator K and parameterize it directly in Fourier space:

$$(Ku_t)(\mathbf{x}) = \mathcal{F}^{-1}(\mathcal{F}(\kappa) \cdot \mathcal{F}(u_t)) = \mathcal{F}^{-1}(R \cdot \mathcal{F}(u_t)), \quad (5)$$

where $R(\mathbf{k}) \in \mathbb{C}^{d_{u_t+1} \times d_{u_t}}$ is a learnable function in spectral domain, $\mathcal{F}$ is the Fourier transform operator, and $\mathbf{k} = (k_1, \dots, k_n) \in \mathbb{N}^n$ corresponds to non-negative frequencies. To reduce the computational complexity of the model, $R(\mathbf{k})$ is nonzero only for $k_j \leq k_{max,j}$, $j = 1, \dots, n$. This step also provides FNO model to be resolution-robust, because $k_{max,j}$ does not depend on input data shape and so the convolutional layer in FNO operates only on the fixed number of harmonics in Fourier domain. The model architecture is given in Fig. 1.

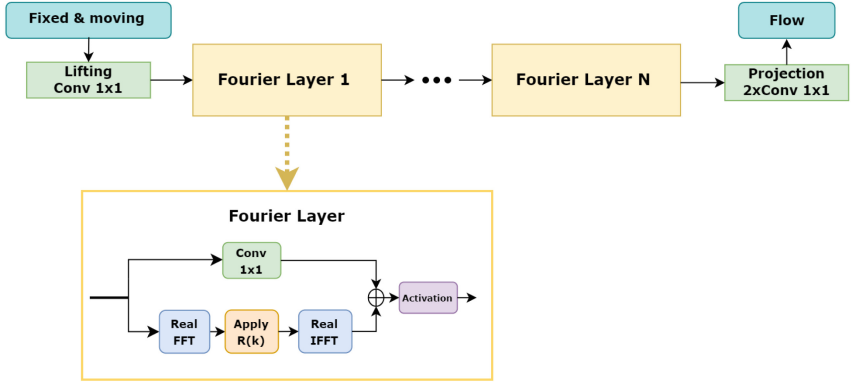


Fig. 1. FNO model architecture. Lifting layer maps the input data into a hidden space and increases the number of channels. Projection layer reduces the number of channel to get the output of the network. Fourier Layer consists the logic from Eq. 3.

2.3 FNOReg Model Architecture

The FNOReg architecture represents an enhancement of the classic FNO architecture designed for image registration. The significance of this improvement will be demonstrated later in Sect. 3.3. The model architecture is depicted in Fig. 2. In comparison to classic FNO [14], our model incorporates several enhancements

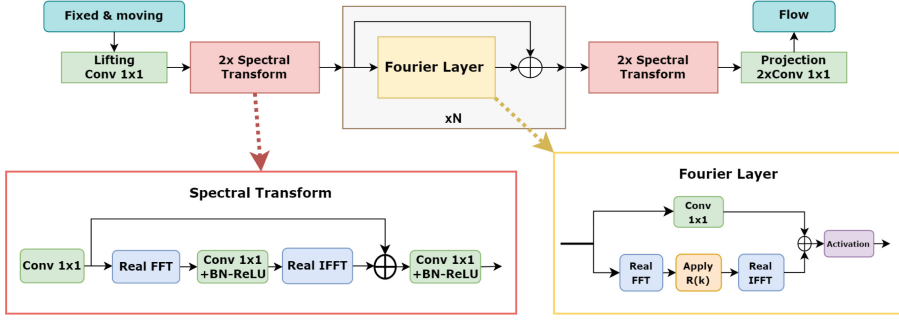


Fig. 2. FNOReg model architecture.

and additions that enable FNOReg to exhibit better quality and increased stability during training.

First, we utilize feature extractors (depicted as red rectangles in Fig. 2) designed with the Spectral Transform layer [7] instead of the standard convolutional layer. The main idea of the Spectral Transform is that convolution in the spectral domain possesses a global receptive field in the spatial domain, allowing us to extract non-local features from our images. This capability enables us to achieve higher registration quality without compromising the model's robustness to input resolution. Our research has shown that incorporating two Spectral Transform layers before and after the sequence of Fourier layers is enough for performance improvement without significantly complicating the model.

Our second improvement involves additional residual connections in the Fourier layers after the activation function. Residual connections tend to stabilize the learning process, and in our experiments FNOReg proved to be more robust when testing different model configurations than standard FNO.

The proposed improvements led to the better learning curve as can be seen in Fig. 3.

2.4 Training Pipeline

The process of training the model for image registration at each iteration includes a forward pass of data through the model, deforming the moving image according to the deformation field, calculating the loss function, and updating the model parameters. First, the fixed image f and moving image m are concatenated in the channel dimension, so the input data consist of an image with 2 channels. After propagating this image through the model, we obtain a deformation field ϕ . Then, we apply this deformation field to the moving image using a spatial transform layer [10], which is based on linear interpolation and implemented as described in [4]. Finally, we update the parameters of the model according to the value of the loss function, which is described below.

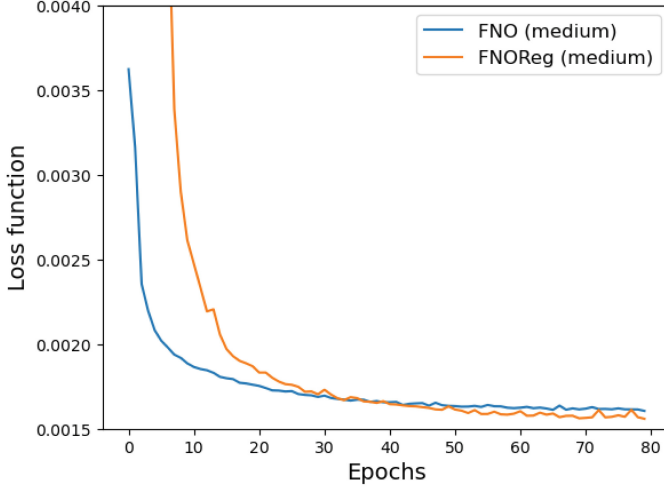


Fig. 3. The loss functions of the FNO and FNOReg models for the OASIS-1 2D dataset. The FNOReg loss starts at a value of 0.85, but its curve is much steeper than that of FNO, as evidenced by the curves intersecting.

Loss Function. We use a common loss function for image registration consisting of two components, $\mathcal{L}_{sim}$ and $\mathcal{L}_{smooth}$:

$$\mathcal{L}(f, m, \phi) = \mathcal{L}_{sim}(f, m_\phi) + \lambda \mathcal{L}_{smooth}(\phi).$$

$\mathcal{L}_{sim}(f, m_\phi)$ penalizes the difference between the fixed and moved images. In our experiments, this component was mean squared error computed as:

$$MSE(f, m_\phi) = \frac{1}{|\Omega_p|} \sum_{p \in \Omega_p} (f(\mathbf{p}) - m_\phi(\mathbf{p}))^2.$$

$\mathcal{L}_{smooth}$ is a regularization term that allows for more realistic and smoother deformation fields. We use a diffusion regularizer [2] implemented with finite difference approximation of gradient. Specifically, the following formula for $\mathcal{L}_{smooth}$ was used:

$$\mathcal{L}_{smooth}(\phi) = \frac{1}{|\Omega_p|} \sum_{p \in \Omega_p} \|\nabla \phi(\mathbf{p})\|^2.$$

For 3D data registration, we incorporate an additional term $\mathcal{L}_{seg}$ into the loss function to control segmentation overlap:

$$\mathcal{L}(f, m, s_f, s_{m_\phi}, \phi) = \mathcal{L}_{sim}(f, m_\phi) + \lambda \mathcal{L}_{smooth}(\phi) + \gamma \mathcal{L}_{seg}(s_f, s_{m_\phi}),$$

where s_f and s_{m_ϕ} represent the anatomical segmentation of f and m_ϕ , respectively. We minimize the Dice loss [17] between s_f^k and $s_{m_\phi}^k$, where k denotes the k -th structure:

$$\mathcal{L}_{seg}(s_f, s_{m_\phi}) = 1 - \frac{2}{K} \sum_{k=1}^K \frac{|s_f^k \cap s_{m_\phi}^k|}{|s_f^k| + |s_{m_\phi}^k|}.$$

To enable automatic differentiation of the loss function, we follow a similar approach to [4], where s_f and s_{m_ϕ} are image volumes with K channels, each channel containing a binary mask defining the segmentation of a specific structure. The warped segmentation s_{m_ϕ} is computed using linear interpolation to ensure differentiability of the loss function.

3 Experiments

3.1 Datasets and Evaluation Metrics

OASIS-1 dataset [16] consists of brain MRI scans from 414 subjects. Each scan includes a segmentation of important anatomical areas. This dataset also provides 414 2D slices and their segmentation masks from corresponding 3D volumes. In our experiments, a preprocessed version of the 2D and 3D OASIS-1 dataset [9] was employed, where all 414 MRI scans were affinely aligned and cropped to the size of 160×192 and $160 \times 192 \times 224$, for 2D and 3D data respectively. For 2D data, we split all scans into 201 for training, 12 for validation, and 201 for testing. After pairing, we obtained 40200 pairs for training, 22 for validation, and 400 for testing. For 3D data, we used 384 scans for training, 10 for validation and 20 for testing; after pairing we ended up with 766 training pairs, 9 validation pairs and 19 test pairs. During the evaluation on the test data, we computed the Dice score for every anatomical area on the fixed and warped images. The evaluation metric for a single pair of images was the average Dice score across all anatomical areas, while the evaluation metric for the entire test dataset was the average across the evaluation metrics for each pair.

3.2 Implementation Details

We implemented our method using the PyTorch [20] framework and neuralop package [14] which is an original implementation of Fourier Neural Operator.

The models were trained as follows:

- For 2D data, we trained our models for 80 epochs using the Adam optimizer with a learning rate of 10^{-4} and a batch size of 8. The parameter λ was set to 0.01.
- For 3D data, we trained our models for 500 epochs using the Adam optimizer with a learning rate of 10^{-4} and a batch size of 1. The parameters λ and γ were set to 0.01.

3.3 Results

We compared FNO and FNOReg with several recent baseline methods based on convolutional neural networks and transformers, which are widely used for biomedical image registration:

1. **Fourier-Net** [11]: A CNN-based model that predicts a band-limited deformation field in the Fourier domain and then obtains the final deformation after zero-padding and inverse DFT.
2. **VoxelMorph** [4]: Model for image registration with a U-Net-based architecture.
3. **VoxelMorph-Large**: A deeper version of VoxelMorph (number of convolution kernels increases in 2 times on each layer, added some convolutional layers between encoder and decoder).
4. **VoxelMorph-Huge** [6]: A customized VoxelMorph model with a comparable parameter size to that of TransMorph that was used in [6] paper.
5. **TransMorph** [6]: A model with a transformer-based encoder and convolutional decoder.

To study the robustness to image resolution, we trained and evaluated all models in two scenarios. In the first scenario, the models were trained and evaluated at full resolution. In the second scenario, the models were trained on resolution reduced by 2 (80×96 pixels for 2D data and $80 \times 96 \times 112$ voxels for 3D data), but evaluated at full resolution, as in the first scenario. For TransMorph, we used the original authors' implementation in [6], and due to technical constraints of the implementation, it was not possible to train this model on halved resolution.

Results on 2D OASIS-1 Data. Table 1 presents the comparison of baseline methods with FNO and FNOReg on the 2D OASIS-1 dataset [16]. We considered three configurations for FNO-based models: small ($d_{u_t} = 16$, $N = 6$, $\mathbf{k}_{max} = (40, 48)$), medium ($d_{u_t} = 32$, $N = 12$, $\mathbf{k}_{max} = (40, 48)$), and large ($d_{u_t} = 32$, $N = 12$, $\mathbf{k}_{max} = (60, 72)$). The best Dice score when training at full resolution is achieved by VoxelMorph-Large, but its quality loss with training resolution reduction is 22.01% (0.777 Dice score versus 0.606). Fourier-Net has fewer parameters than other baseline models (except classical VoxelMorph, which achieved a comparable to Fourier-Net performance with only 91000 parameters) and achieves a 0.757 and 0.761 Dice score (for models with 16 and 32 channels, respectively) in the first training scenario, losing about 20–25% of quality in the second scenario. At the same time, the results on full resolution for FNO-based models are comparable to the results of TransMorph and VoxelMorph-Huge (0.775 DSC for large FNOReg and the same result for TransMorph), but the maximum loss of quality for training on downsampled images for FNO and FNOReg is 0.67% and 0.78%, respectively. For 2D data, the mean inference times were from 0.006 to 0.014s for baseline models, and from 0.011 to 0.021s for FNO-based models.

Table 1. Comparison of the models on the 2D OASIS-1 dataset; 1x and 0.5x indicate training on full and halved resolution, respectively. The Dice score is used for comparison (the bigger the better). The best values in each column of each group (baseline and FNO-based models) are highlighted in bold. The change in Dice score when training on halved resolution compared to full resolution is indicated in the last column.

Model name	#params	Dice on 1x	Dice on 0.5x	Metric change on 0.5x
Initial (only affine)	-	0.544	0.544	-
Fourier-Net (16 channels)	1.4M	0.757	0.611	-19.29%
Fourier-Net (32 channels)	6.1M	0.761	0.571	-24.97%
VoxelMorph	91K	0.760	0.715	-5.92%
VoxelMorph-Large	5.8M	0.777	0.606	-22.01%
VoxelMorph-Huge	21.1M	0.771	0.698	-9.47%
TransMorph	31.0M	0.775	-	-
FNO (small)	1.3M	0.751	0.746	-0.67%
FNOREg (small)	1.3M	0.753	0.751	-0.27%
FNO (medium)	10.0M	0.766	0.762	-0.52%
FNOREg (medium)	10.0M	0.769	0.763	-0.78%
FNO (large)	22.8M	0.769	0.766	-0.39%
FNOREg (large)	22.8M	0.775	0.772	-0.39%

In Fig. 4, the deformation fields obtained by the compared models and their spectra are depicted. From the visualization of deformation fields and their spectra, it can be observed that VoxelMorph and TransMorph deformation fields contain high-frequency modes, corresponding to the vertical cross on the spectra. The FNO-based models and Fourier-Net produce smoother deformation fields. However, in Fourier-Net, the spectrum is explicitly filtered with a box-shaped low-pass filter inside the model, which theoretically produces some aliasing and ringing artifacts that may affect the quality of the deformation fields.

Results on 3D OASIS-1 Data. Table 2 presents a comparison of baseline methods with FNOREg on the 3D OASIS-1 dataset [16]. Due to computational constraints, we considered only one configuration of the FNOREg model ($d_{u_t} = 12$, $N = 15$, $\mathbf{k}_{max} = (40, 48, 56)$). For 3D data, obtaining a reasonable hyperparameter configuration to train the standard FNO models was not feasible, as the training process did not converge. This highlights the significance of the improvements presented in the proposed FNOREg model. While TransMorph achieved the best Dice score when trained at full resolution, it was not possible to train it on half resolution due to limitations in the authors' implementation. VoxelMorph-Large attained the second-best result, but its quality degraded by 24.68% (0.863 Dice score versus 0.650) with training resolution reduction. Fourier-Net exhibited a similar quality loss as for 2D data when trained on half resolution. In contrast, the standard VoxelMorph model demonstrated a more significant quality decrease when trained on reduced resolution for 3D data com-

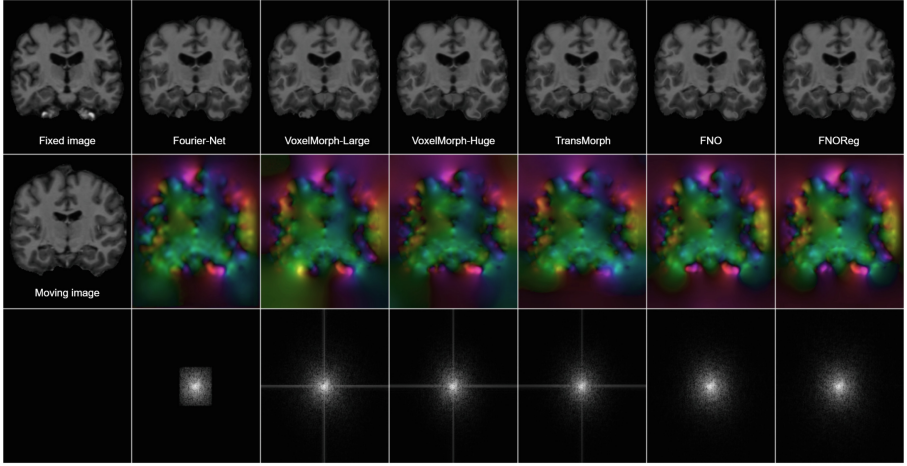


Fig. 4. Visual comparison of different methods on 2D OASIS-1 dataset. First column consists of fixed image and moving image. Other columns consist (from top to bottom) of moved image obtained from method, deformation field visualization, and deformation field DFT amplitude.

pared to 2D data. FNOReg also showed a slightly larger quality decrease when trained on downsampled images compared to 2D data (2.65%), but it remained considerably less than the other models. For 3D data, the mean inference times were from 0.087 to 1.048 s for baseline models, and 1.075 s for FNOReg.

Smoothness of Deformation Fields. To evaluate the smoothness of deformations obtained from different models, we calculate the percentage of voxels with a negative value of the Jacobian determinant of the deformation field (so-called “folded voxels” [6]) for each model. The results of this evaluation are presented in Table 3 for 2D data and in Table 4 for 3D data. As one can see, Fourier-Net achieves the best result at the original resolution due to the explicit low-pass filtering of the deformation field. However, in the case of Fourier-Net, the number of folded voxels strongly increases at halved resolution, leading to a degradation in performance. The smoothness of deformations from VoxelMorph and VoxelMorph-Huge slightly changes at halved resolution, but the absolute values of the metric at the original resolution for these models are worse than for other models (especially on 3D data). The FNOReg model at full resolution achieves the second-best result on 2D data and the forth-best result on 3D data. Moreover, the number of folded voxels in deformations obtained by FNOReg increases only slightly at halved resolution. To summarize, the FNOReg model

Table 2. Comparison of the models on the 3D OASIS-1 dataset; 1x and 0.5x indicate training on full and halved resolution, respectively. The Dice score is used for comparison (the bigger the better). The best values in each column of each group (baseline and FNO-based models) are highlighted in bold. The change in Dice score when training on halved resolution compared to full resolution is indicated in the last column.

Model name	#params	Dice on 1x	Dice on 0.5x	Metric change on 0.5x
Initial (only affine)	-	0.572	0.572	-
Fourier-Net (16 channels)	4.5M	0.814	0.617	-24.20%
Fourier-Net (32 channels)	17.8M	0.822	0.619	-24.70%
VoxelMorph	274K	0.834	0.737	-11.63%
VoxelMorph-Large	15.2M	0.863	0.650	-24.68%
VoxelMorph-Huge	63.3M	0.849	0.722	-14.96%
TransMorph	46.8M	0.867	-	-
FNOReg	98.1M	0.830	0.808	-2.65%

has the best absolute values for the folded voxels metric among the methods that do not show significant degradation in the deformation field smoothness at halved resolution. In other words, it achieves a tradeoff between smoothness of deformation at the original resolution and only its minor degradation at reduced resolution.

Table 3. Comparison of the smoothness of deformations from different models on the 2D OASIS dataset. A lower metric value indicates better performance. The best values in each column of each group (baseline and FNO-based models) are highlighted in bold.

Model name	% of $ J_\phi < 0$ on 1x	% of $ J_\phi < 0$ on 0.5x
Fourier-Net (16 channels)	0.668 $\pm$ 0.353	6.802 $\pm$ 1.499
Fourier-Net (32 channels)	0.751 $\pm$ 0.374	7.477 $\pm$ 1.573
VoxelMorph	0.717 $\pm$ 0.368	0.743 $\pm$ 0.365
VoxelMorph-Large	0.744 $\pm$ 0.371	7.213 $\pm$ 1.632
VoxelMorph-Huge	0.726 $\pm$ 0.366	0.512 $\pm$ 0.264
TransMorph	0.704 $\pm$ 0.365	-
FNO (large)	0.718 $\pm$ 0.356	0.802 $\pm$ 0.389
FNOReg (large)	0.690 $\pm$ 0.355	0.767 $\pm$ 0.370

Table 4. Comparison of the smoothness of deformations from different models on the 3D OASIS dataset. A lower metric value indicates better performance. The best values in each column of each group (baseline and FNO-based models) are highlighted in bold.

Model name	% of $ J_\phi < 0$ on 1x	% of $ J_\phi < 0$ on 0.5x
Fourier-Net (16 channels)	0.167 $\pm$ 0.054	3.365 $\pm$ 0.357
Fourier-Net (32 channels)	0.213 $\pm$ 0.064	2.687 $\pm$ 0.326
VoxelMorph	0.772 $\pm$ 0.127	0.735 $\pm$ 0.131
VoxelMorph-Large	0.203 $\pm$ 0.047	3.197 $\pm$ 0.391
VoxelMorph-Huge	0.726 $\pm$ 0.123	0.717 $\pm$ 0.100
TransMorph	0.680 $\pm$ 0.118	-
FNOReg	0.314 $\pm$ 0.063	0.572 $\pm$ 0.080

4 Conclusion

This paper introduces FNOReg, a novel model for image registration that builds upon the original FNO architecture. Using the properties of Fourier neural operators, FNOReg offers enhanced robustness to input data resolution. Our model incorporates two significant improvements over the original FNO: the integration of feature extractors based on the Spectral Transform layer [7] and the utilization of additional residual connections in the Fourier layers. Moreover, we tailored the training pipeline specifically for the image registration problem. In addition to improving registration quality, our enhancements to the FNO architecture stabilize the training process and enable the model to be trained in cases where the standard FNO fails.

Through comprehensive evaluations on the OASIS-1 2D and 3D datasets [16], we found that FNOReg achieves comparable quality to popular models widely used for image registration, such as VoxelMorph [4], TransMorph [6], and FourierNet [11]. Notably, our FNOReg models maintain consistent performance even when trained on data with reduced resolutions, unlike other models that exhibit a significant decrease in registration quality. This aspect of our method is particularly advantageous for reducing memory consumption when handling large datasets and can be especially beneficial for 3D data. Additionally, our comparison of deformation spectra obtained from different models reveals that FNO and FNOReg yield smoother deformation fields compared to their counterparts.

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